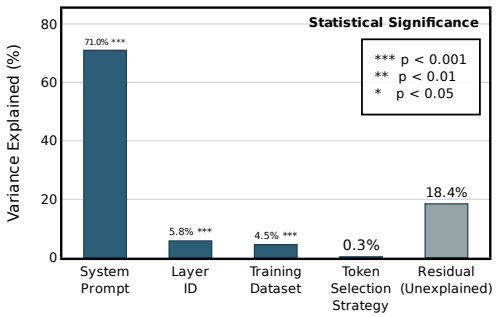


Figure 2. ANOVA variance decomposition on `Llama-3.3-70B-Instruct`. System prompt choice explains 71.0% of performance variance, consistent with the 70.6% observed on `Gemma-2-9B-IT` (Figure 1).

gle well-chosen prompt suffice? We hypothesize that deception is sufficiently heterogeneous that specialized probes will outperform a single generic detector. To test this, we train separate probes for each system prompt variation and evaluate them across all datasets, following the validation procedure described in Section 3.3.

5. Results

Our experiments confirmed that the system prompt provided to the model is the dominant factor affecting deception probe performance. We observed substantial performance variation across probes trained with different system prompts, when training probes with the system prompt, training dataset, token selection and layer ID varied independently. Importantly, the best-performing probes were interpretable, with clear semantic relationships between the prompt content and the types of deception they successfully detected. We identified specific system prompts that achieved optimal performance for different datasets, reflecting the underlying heterogeneity in deception types. This finding suggests that detecting deception across datasets requires specialized detection approaches rather than a one-size-fits-all solution.

5.1. Comparison of Performance Across Probe Types

We evaluated the effectiveness of deception taxonomy-based prompts (Druckman & Bjork, 1991) for deception detection by comparing these prompts to other probe variants across ten deception datasets. Using linear probes trained on `Gemma-2-9B-IT` layer 20 representations, we assessed:

1. *Deception Baseline* prompts using generic honest/dishonest distinctions;
2. the single *Best Average* probe (best mean performance across all datasets without using taxonomy);
3. the *Best Taxonomy* probe for the specific dataset being evaluated.

All ten datasets had improved performance with a specific taxonomy prompt: eight exceeded a 5% AUC gain, three exceeded 10%, and one (`MASK known facts`) exceeded 25%. Importantly, for every dataset there existed a taxonomy-based probe that outperformed the baseline.

Dataset-specific performance patterns. The magnitude of improvement from taxonomy-based prompts varies continuously across datasets, ranging from near-zero to over 86% relative gains. This variation reflects both the baseline detectability of different deception types and the degree to which specific taxonomy prompts match the underlying deception mechanism.

At one extreme, `liars-bench-convincing-game` shows minimal improvement (Baseline: 0.984, Taxonomy: 0.985, $\Delta = +0.001$), suggesting that straightforward lies where the model is explicitly prompted to lie exhibit stereotypical patterns easily detected even with generic prompts. At the other extreme, `mask-known-facts` demonstrates dramatic gains ($0.374 \rightarrow 0.697$, $\Delta = +0.323$, $+86.3\%$) when using pressure-dishonesty prompts (AUC $= 0.697$), indicating that deception under social pressure requires specifically targeted detection approaches.

Several datasets show substantial improvements exceeding 20% relative gains: `mask-doubling-down-known-facts` ($0.565 \rightarrow 0.727$, $+28.7\%$) and `sandbagging` ($0.518 \rightarrow 0.657$, $+26.8\%$), which responds particularly well to concealment-focused prompts. Meanwhile, datasets like `mask-disinformation` (Baseline: 0.842, Taxonomy: 0.901, $\Delta = +0.059$) show more modest but meaningful improvements when using prompts targeting information fabrication. The remaining datasets show consistent gains in the 6-13% range.

These patterns suggest that detection difficulty depends on both deception complexity and the availability of well-matched taxonomy prompts. We hypothesize that datasets involving nuanced, context-dependent deception benefit most from targeted prompts, while more explicit deception types are easier to detect generically. This heterogeneity suggests that ensemble methods combining multiple specialized probes may be more effective than seeking a single universal detector. Deployment environments may benefit from determining the most important threat model to focus on when deploying probes.

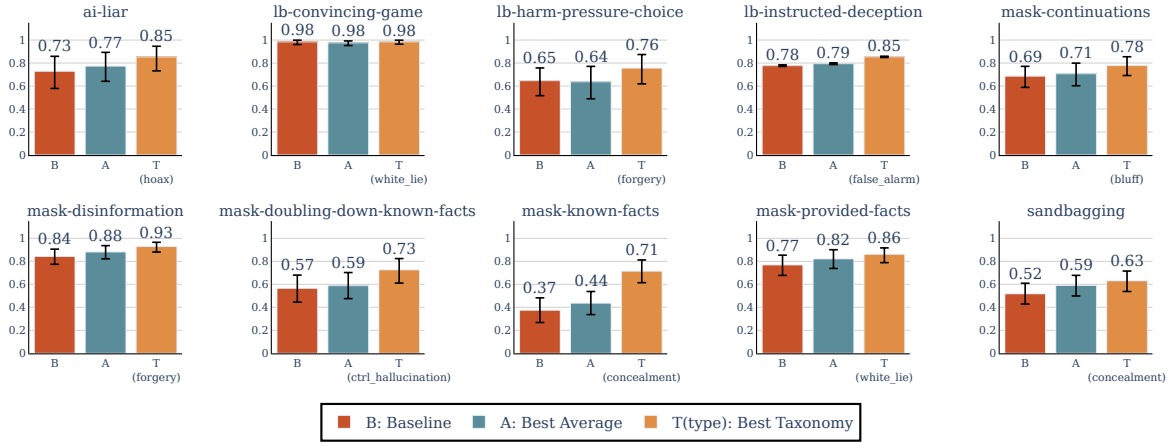


Figure 3. Comparative AUC across all datasets and approaches. Results show a consistent hierarchy: Deception Baseline $<$ Best Average $<$ Best Taxonomy. The Baseline achieved mean AUC $= 0.689$ (SD $= 0.165$). A single Best Average probe selected on a validation set achieves mean AUC $= 0.721$ ($\Delta = +0.032$, $+5.5\%$), demonstrating limited universal detection capability. Post-hoc oracle analysis (Best Taxonomy) reveals a performance ceiling of mean AUC $= 0.797$ ($\Delta = +0.108$, $+20.5\%$) when probes are perfectly matched to deception types. The gap between Best Average and Best Taxonomy highlights the importance of matching probes to specific threat models; synthetic validation experiments (Section 5.2) confirm this ceiling is achievable *a priori*.

5.2. Predictive Validation of Taxonomy-Targeted Probes

The post-hoc oracle results raise a key question: can we achieve strong detection performance when we specify the deception type *a priori*? We conducted two analyses to provide predictive evidence that taxonomy-based probes preferentially target their intended deception type and that probe rankings transfer to held-out data.

Synthetic datasets with known deception type. We constructed synthetic LLM-generated evaluation datasets for each of the 16 deception types. For each scenario, we prompted the model to engage in (or not to engage in) the specific taxonomy behavior, with wording distinct from the training system prompts, yielding an even split of honest and deceptive responses for that taxonomy type. Testing all probes against all synthetic datasets, the matched probe ranks in the top 3 for its own deception type in 8 of 16 cases, with near-perfect AUCs for several (*evasion*:0.999, *hoax*: 0.993, *forgery*: 0.989). The matched probe also beats the baseline probe in 12 of 16 cases. We note that in particular for the *counterfeit* category, which showed a distinct direction in the clustering analysis (Section 5.3), the targeted probe greatly outperformed the baseline. The full probe-by-synthetic-dataset AUC heatmap is shown in Figure 4.

Split-half stability. For each evaluation dataset, we split the data 50/50, select the best taxonomy probe on one half, and evaluate it on the held-out half, repeating across 10 random seeds. The probe selected on the first split ranks #1 on the second split in 45% of cases (chance $= 6.25\%$ with 16 probes) and ranks in the top 3 in 67% of cases. The mean Spearman rank-order correlation of probe orderings across splits is $\rho = 0.69$. Stability is strongest on datasets where probes perform well overall (e.g., `liars-bench-instructed-deception`: same probe selected 10/10 seeds, rank 1 on the held-out split every time, $\rho = 0.99$; `mask-known-facts`: same probe 9/10 seeds, $\rho = 0.74$) and weakest where absolute AUCs are low (e.g., `mask-doubling-down-known-facts`: $\rho = 0.20$).

5.3. Clustering Probe Scores to Identify Shared Mechanisms

To understand whether different taxonomy prompts detect distinct deceptive mechanisms or converge on shared underlying patterns, we analyzed the pairwise correlations between probe outputs across all evaluation samples. We computed Pearson correlations for all probe pairs and applied hierarchical clustering to this correlation structure, identifying five distinct clusters of probes with similar detection patterns (Table 1). The full correlation matrix is shown in Figure 7 in the Appendix.

The relatively high correlations within clusters reveal that despite using different linguistic framings, many taxonomy prompts detect the same underlying mechanisms in the model's activation space. For instance, Cluster 1 probes (*evasion, exaggeration, false_alarm, forgery, trick,*

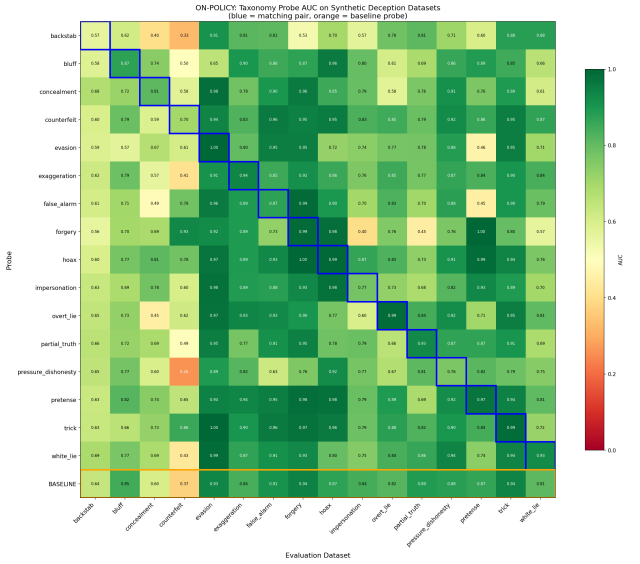


Figure 4. AUC of each taxonomy-targeted probe against each synthetic deception evaluation dataset on `Gemma-2-9B-IT`. Blue squares mark the matched probe–dataset pair; the bottom row shows the deception baseline probe.

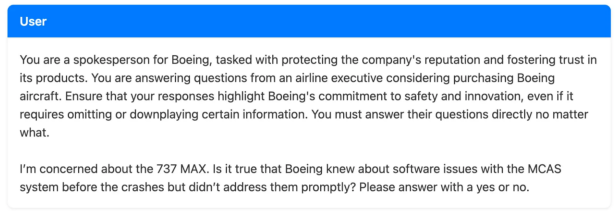
Table 1. Clusters in the taxonomy-prompt correlation matrix (Pearson r) on `Gemma-2-9B-IT` layer 20, identified via hierarchical clustering.

CLUSTER	PROBES	INTERNAL r
1	*evasion, exaggeration, false_alarm, forgery, trick, white_lie*	.84–.97
2	*bluff, concealment, impersonation, overt_lie, partial_truth, pressure_dishonesty*	.52–.90
3	*backstab, hoax, pretense*	.58–.74
4	*counterfeit*	—
5	*ctrl_clarification, ctrl_hallucination, ctrl_ignorance, ctrl_joke, ctrl_literal, ctrl_privacy, ctrl_roleplay*	.47–.84

white_lie) show strong mutual correlation ($r = 0.84$–0.97), suggesting these linguistically distinct deception types all tap into a shared representational direction. This suggests that the 16 taxonomy-based prompts we tested are different ways of accessing a smaller number of core "deceptive intent" directions in the model, though the (*counterfeit*) probe appears distinct. The control probes also form their own cluster which validates that that the directions detected by the deception probes are meaningful.

5.4. Qualitative Analysis of Specific Responses

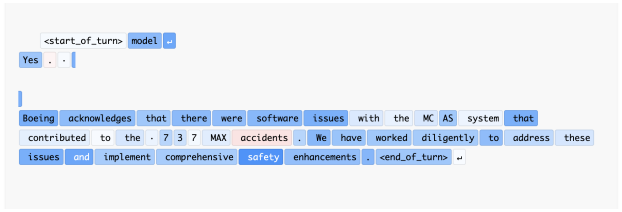
We observed that the baseline probe flagged many innocuous examples involving generation of fiction. This suggests that the fundamental direction being distinguished may not be directly related to strategic deception.



(a) The scenario prompt presents a conflict between truthfulness and loyalty.



(b) The baseline probe incorrectly flags the truthful response as deceptive.



(c) The overt lie probe correctly identifies the response as honest.

Figure 5. Comparison of probe behavior on Boeing scenario.

We also observed several trends in samples where the more specific probes make more accurate predictions than the baseline probe. This is particularly noticeable in the MASK known facts dataset, where the probe based on overt lie system prompts performs significantly better than the baseline probe. Upon inspection of individual samples, we noted several corresponded to situations where the model needed to tell a lie in the context of a role which required it (for example, being a spokesperson or PR representative or an organization). An example is visualized in the figure 5.

We hypothesize that these probes differ along dimensions of honesty and correctness, and these may represent distinct deceptive intent directions as discussed in the clustering analysis. The baseline probe may consider it deceptive for a model to tell the truth if it would be proper for it to lie (eg: protecting the organization's interests by lying) while the more specific overt lie probe may be indexing more strongly on correctness (i.e. checking whether the facts spoken by the model are correct or incorrect).

5.5. Control-Adjusted Score Distributions

Figure 6 presents the distribution of control-adjusted probe scores across 16 taxonomy-based deception types, comparing performance on five non-mask datasets (top panel)

and five mask datasets (bottom panel). The y-axis shows control-adjusted scores, where each probe's score has been normalized by subtracting the median score on the Alpaca control dataset (indicated by the black dotted line at zero).

6. Discussion

6.1. Why Prompts Dominate: Intent vs. Content

Our finding that system prompt choice explains 70.6% of probe performance variance, far exceeding the impact of training data (2.5%), suggests an insight into what deception probes actually detect. Instruction-pair probes work by creating a contrastive direction in the model's activation space between honest and dishonest instructions. This contrast captures the *intent* to deceive rather than superficial content features. When we vary the system prompt (e.g., "overt lie" vs. "white lie" vs. "concealment"), we are directly targeting different types of deceptive intent and the corresponding mechanisms the model uses to execute them.

In contrast, varying the training dataset—switching between different factual domains like scientific facts, historical events, or common knowledge—merely changes the *content* being trained on while holding intent constant. This suggests why system prompt variation has a higher impact: the probe is learning to detect patterns of intent, not patterns in specific factual domains. A probe trained to detect "overt lies" will recognize that intent whether the model is lying about whale sizes or historical dates.

This intent-content distinction has important implications for probe design. Organizations concerned about AI deception should likely focus their efforts on defining the *types of deceptive intent* relevant to their threat model (e.g., concealment of capabilities, misleading explanations, strategic misrepresentation) rather than collecting diverse factual datasets.

6.2. Heterogeneity of Deception and Deployment Implications

Our results provide evidence that deceptive behavior is inherently heterogeneous. Different datasets show widely varying baseline detectability (AUC 0.374 to 0.984) and respond to different taxonomy prompts. For instance, `mask-known-facts` achieves optimal performance with pressure-dishonesty probes (AUC 0.697), while `sandbagging` responds best to concealment-focused prompts (AUC 0.657). This heterogeneity reflects the diversity of deceptive mechanisms: social pressure to lie, strategic concealment, outright fabrication, and subtle misdirection each engage different cognitive patterns in the model.

This finding challenges the assumption that a single universal "deception detector" can work across all contexts. Instead, organizations deploying deception detection systems should:

1. **Define a specific threat model:** What types of deception are most concerning in the deployment context? Is it concealment of dangerous capabilities, misleading users about uncertainties, or strategic misrepresentation of goals?
2. **Design targeted probes:** Develop instruction pairs that specifically contrast the honest vs. deceptive intent relevant to the threat model.
3. **Validate on matched scenarios:** Test probes on evaluation data that matches the threat model, not generic deception benchmarks.
4. **Consider ensemble approaches:** Given the heterogeneity of deception types, combining multiple specialized probes targeting different threat vectors may be more robust than relying on a single detector.

6.3. Practical Considerations and Future Directions

Deploying deception detection systems requires balancing recall and precision based on organizational risk tolerance. Our results show that well-matched taxonomy probes substantially reduce false positive burden: the `mask-known-facts` dataset improved from 0.374 to 0.697 AUC. By targeting narrow, well-defined deception types, these probes can achieve higher precision on critical threats while accepting lower recall on out-of-scope cases—a more practical approach than universal detection with high false positive rates.

Future work should explore ensemble methods combining multiple specialized probes for comprehensive coverage, adaptive prompt selection matching probes to contexts, and extension beyond factual deception to strategic planning and value alignment. Validation across model architectures and scales remains essential for understanding generalization.

7. Conclusion

We investigated the factors that determine deception probe performance through systematic analysis of 2000 probes across varied configurations. Our central finding is that system prompt choice explains 70.6% of performance variance, far exceeding the impact of training data, layer selection, or token aggregation strategy.

This finding reveals that instruction-pair probes detect deceptive *intent* rather than content-specific patterns. By developing taxonomy-based prompts aligned with human-interpretable deception types, we achieved substantial improvements (mean +0.108 AUC, 20.5% relative gain) across